

How to use the Dataplex data lineage feature

So far, you've learned that data tracing is the practice of tracking metadata to provide insight into the path data has taken through an entire data lifecycle. Dataplex's lineage feature provides a built-in data tracing tool for all the data in your organization. This means you can review and search metadata for available data across your organization, and trace its transformation as it moves through the data lifecycle. In this reading, you'll learn how to use the Dataplex data lineage feature.

Enabling data lineage tracing

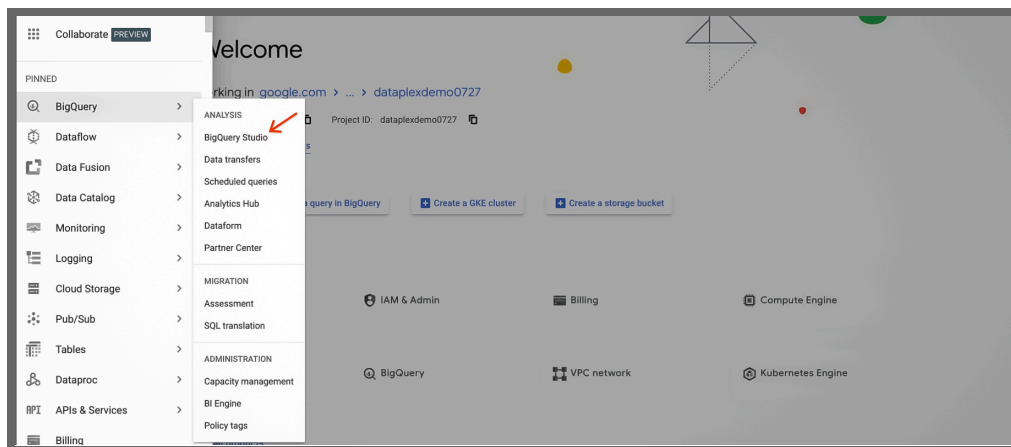
To use the automatic lineage tracing features offered by Google Cloud, you need to enable the Data Lineage API in Dataplex. The Data Lineage API traces metadata from BigQuery, Cloud Data Fusion, Cloud Composer, and Dataproc.

Metadata collected from BigQuery includes Copy, Load, and Query jobs. Specific query jobs include creating tables and views. Also, SQL commands like SELECT, MERGE, UPDATE, and DELETE can be tracked with Data Lineage API.

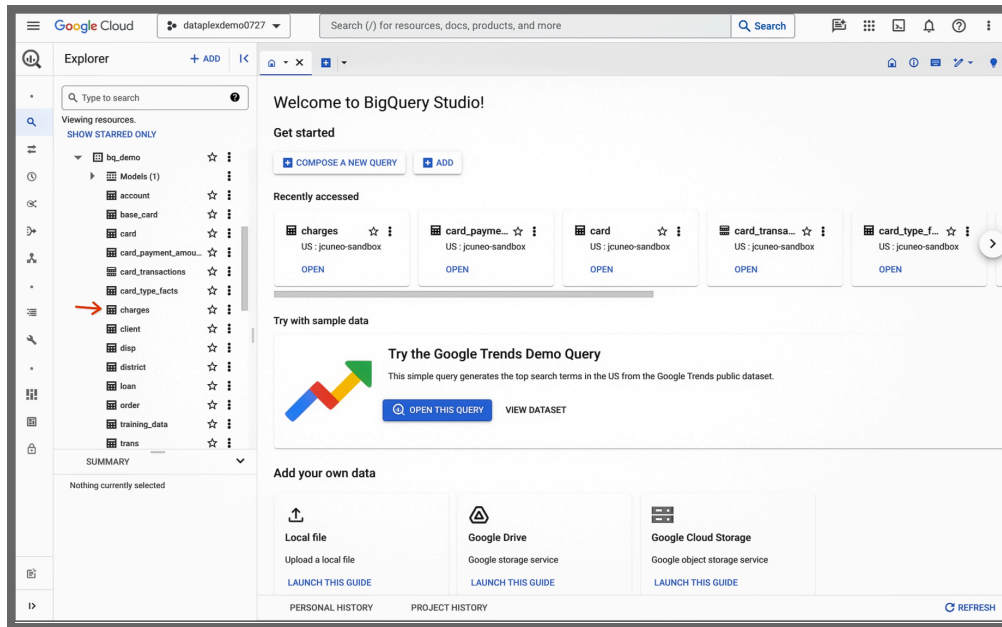
Navigate to the data lineage graph in BigQuery

To access the data lineage graph, first navigate to BigQuery from the Google Cloud console. Then within BigQuery, navigate directly to the data lineage graph with these steps:

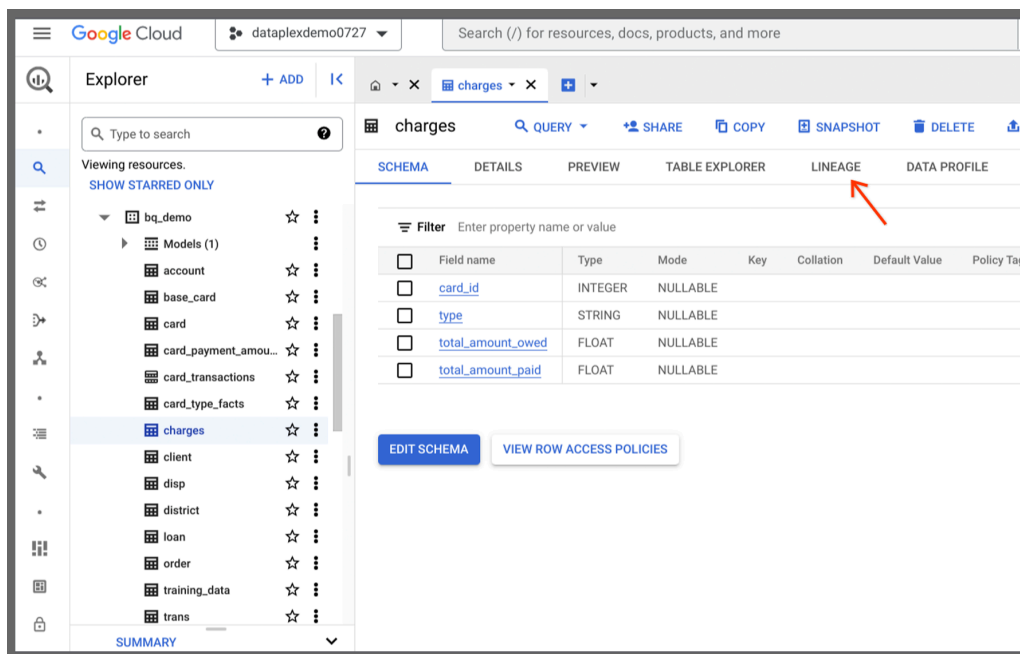
1. Open the BigQuery SQL workspace page.



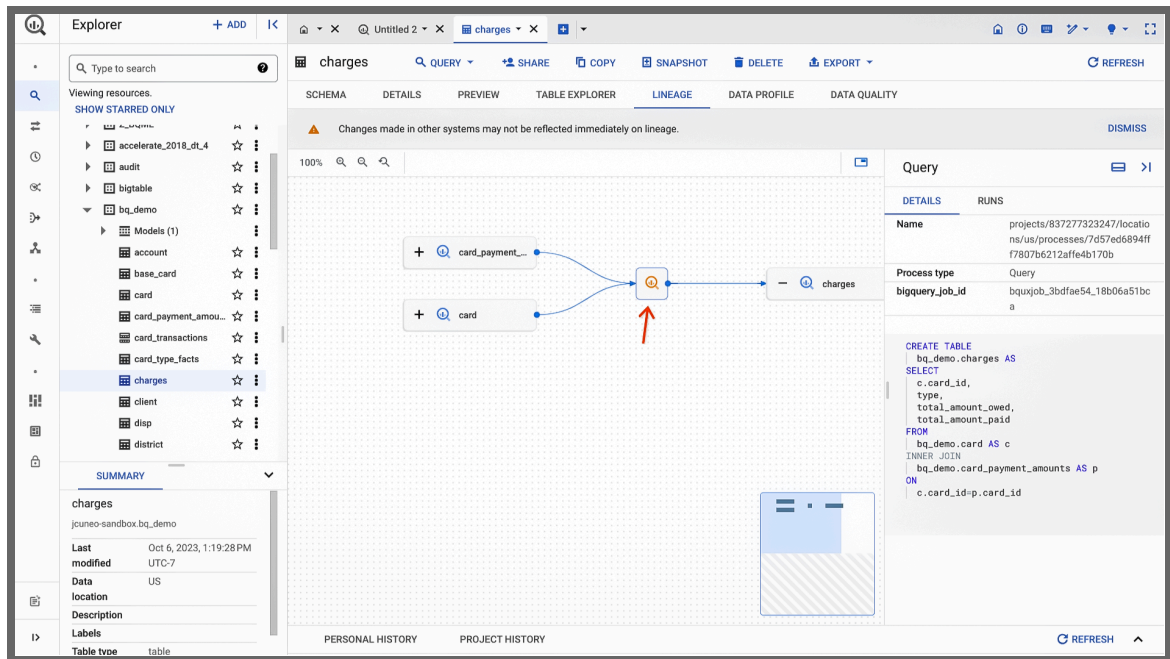
- Open the preferred table to review the data lineage.



- Click the “Lineage” tab.



- Select each of the process buttons to learn more about the transformation or action that occurred.

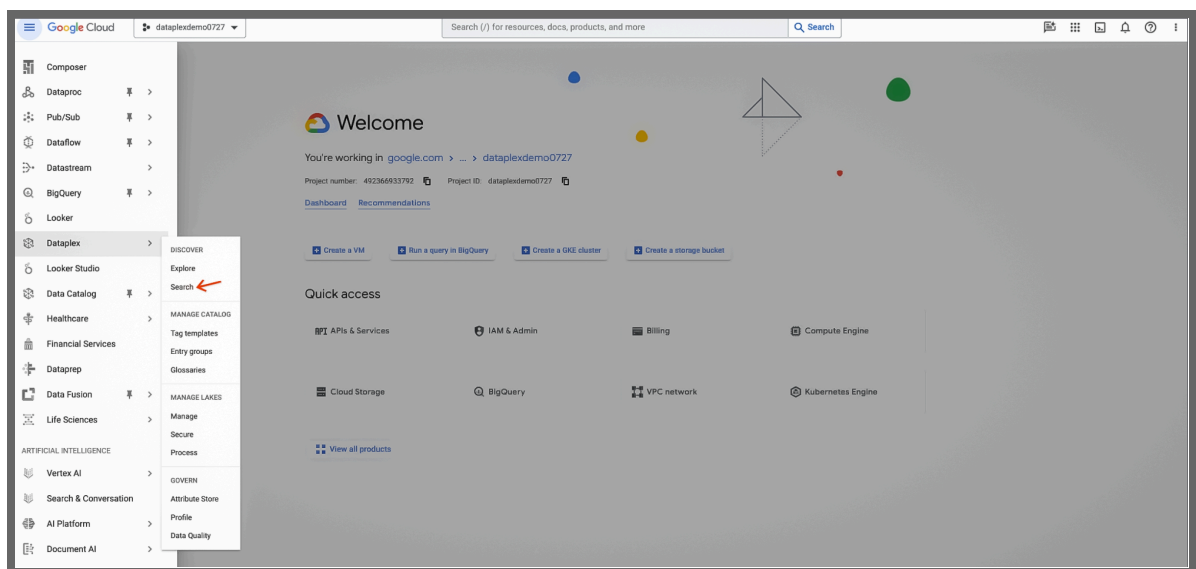


To learn more information about the action, click any of the BigQuery icons to reveal a “Details” table.

Navigate to the data lineage graph in Dataplex

Another way to access the data lineage graph is to navigate to Dataplex from the Google Cloud console. In the Dataplex user interface, you can navigate directly to the data lineage graph by completing these steps:

1. Open the Dataplex search page.



2. Navigate to the entry details page.

The screenshot shows the Google Cloud Dataplex console interface. The left sidebar contains navigation options like Discover, Explore, Search, Manage Catalog, Tag templates, Entry groups, Glossaries, Manage Lakes, Manage, Secure, Process, Govern, Attribute Store, Profile, and Data Quality. The main area displays search results for 'charges'. A red arrow points to the 'charges' entry in the list. The entry details show it is a BigQuery table named 'charges' in the 'unvalidated_service_billingty.com' dataset, created on Oct 6, 2023, at 1:19:28 PM. The table description is 'Marketplace customer refunds data. For each partition date, it only includes refund records completed on the day.'

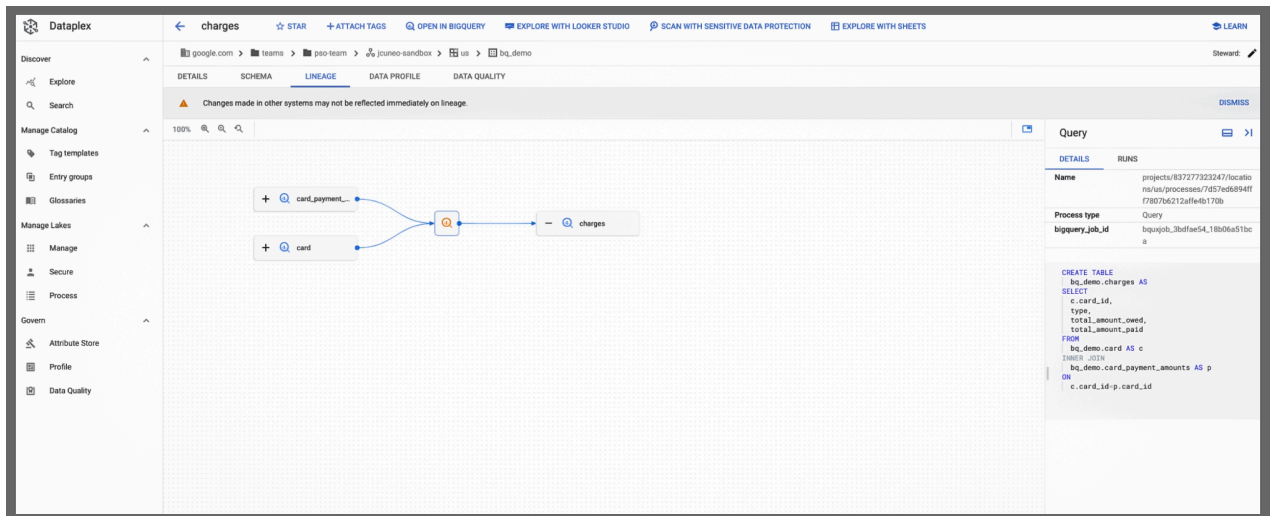
Dataset	Table	System	Location	Resource URL	Labels	Fully qualified name	Description
unvalidated_service_billingty.com	charges	BigQuery	us	juneeo-sandbox.bq_demo.charges	Edit in BigQuery	bigquery:juneeo-sandbox.bq_demo.charges	Marketplace customer refunds data. For each partition date, it only includes refund records completed on the day.

3. Click the “Lineage” tab.

The screenshot shows the Google Cloud Dataplex console interface with the 'charges' table selected. The 'LINEAGE' tab is active, as indicated by a red arrow. The 'BigQuery table details' section shows the table's metadata, including its type (TABLE), system (BIGQUERY), and location (us). The 'Overview' section shows the table's creation time (Oct 6, 2023, 1:19:28 PM) and last modification time (Oct 6, 2023, 1:19:28 PM). The 'Tags' section shows that there are no tags attached to this data asset.

Property	Value
Type	TABLE
System	BIGQUERY
Table type	BigQuery table
Creation time	Oct 6, 2023, 1:19:28 PM
Last modification time	Oct 6, 2023, 1:19:28 PM
Expiration time	Never

4. Open a data lineage diagram with icons indicating which service transformed the data .



Key takeaways

BigQuery and Dataplex offer platform-native options to generate data lineage graphs. Data that has been queried by BigQuery or managed by Dataplex comes with built-in tracking for data events.

Resources for more information

For source documentation about how to create a data lineage graph, check out these links:

- Google Cloud documentation provides step-by-step directions on how to create a data lineage graph:

<https://cloud.google.com/data-catalog/docs/how-to/lineage-gcp#view-bq-lineage-graphs>